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A Standard flat_map

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0.1 Revisions

0.1.1 Changes from R5

- Address comments from San Diego meeting LWG small group review.
- Replace *Requires:* elements with new-style elements.
- Remove `merge()` member functions, based on LEWG guidance.
- Numerous other corrections.

0.1.2 Changes from R4

- Address comments from Batavia meeting review.

0.1.3 Changes from R3

- Remove previous sections.
- Retarget to LWG exclusively.
- Wording.

0.1.4 Changes from R2

- `value_type` is now `pair<const Key, T>`.
- `ordered_unique_sequence_tag` is now `sorted_unique_t`, and is applied uniformly such that those overloads that have it are assumed to receive sorted input, and those that do not have it are not.
- The overloads taking two allocators now take only one.
- `extract()` now returns a custom type instead of a `pair`.
- Add `contains()` (tracking `map`).

0.1.5 Changes from R1

- Add deduction guides.
- Change `value_type` and reference types to be proxies, and remove `{const_}, pointer`.
- Split storage of keys and values.
- Pass several constructor parameters by value to reduce the number of overloads.
- Remove the benchmark charts.

0.1.6 Changes from R0

- Drop the requirement on container contiguity; sequence container will do.
- Remove `capacity()`, `reserve()`, and `shrik_to_fit()` from container requirements and from `flat_map` API.
- Drop redundant implementation variants from charts.
- Drop erase operation charts.
- Use more recent compilers for comparisons.
- Add analysis of separated key and value storage.

21 Containers library

[containers]

21.1 General

[containers.general]

- ¹ This Clause describes components that C++ programs may use to organize collections of information.
- ² The following subclauses describe container requirements, and components for sequence containers and associative containers, as summarized in Table 76.

Table 1 — Containers library summary

Subclause	Header(s)
21.2 Requirements	
21.3 Sequence containers	<array> <deque> <forward_list> <list> <vector>
21.4 Associative containers	<map> <set>
21.5 Unordered associative containers	<unordered_map> <unordered_set>
21.6 Container adaptors	<queue> <stack> <flat_map>
21.7 Views	

21.2.3 Sequence containers

[sequence.reqmts]

- ¹ A sequence container organizes a finite set of objects, all of the same type, into a strictly linear arrangement. The library provides four basic kinds of sequence containers: **vector**, **forward_list**, **list**, and **deque**. In addition, **array** is provided as a sequence container which provides limited sequence operations because it has a fixed number of elements. The library also provides container adaptors that make it easy to construct abstract data types, such as **stacks**, **queues**, [flat_maps](#), or [flat_multimaps](#), out of the basic sequence container kinds (or out of other kinds of sequence containers).

21.2.6 Associative containers

[associative.reqmts]

- ¹ Associative containers provide fast retrieval of data based on keys. The library provides four basic kinds of associative containers: **set**, **multiset**, **map** and **multimap**. [The library also provides container adaptors that make it easy to construct abstract data types, such as flat_maps or flat_multimaps, out of the basic sequence container kinds \(or out of other program-defined sequence containers\).](#)

- ⁶ ~~iterator of an associative container is of the bidirectional iterator category.~~ [An associative container's iterator meets the bidirectional iterator requirements.](#) For associative containers where the value type is the same as the key type, both **iterator** and **const_iterator** are constant iterators. It is unspecified

whether or not `iterator` and `const_iterator` are the same type. *Remark:* `iterator` and `const_iterator` have identical semantics in this case, and `iterator` is convertible to `const_iterator`. Users can avoid violating the one-definition rule by always using `const_iterator` in their function parameter lists.

21.6 Container adaptors

[container.adaptors]

21.6.1 In general

[container.adaptors.general]

- ¹ The headers `<queue>` ~~and~~ `<stack>`, and `<flat_map>` define the container adaptors `queue`, `priority_queue` ~~and~~ `stack`, and `flat_map`.
- ² For container adaptors, no `swap` function throws an exception unless that exception is thrown by the `swap` of the adaptor's `Container`, `KeyContainer`, `MappedContainer`, or `Compare` object (if any).
- ³ For container adaptors that have them, the `insert`, `emplace`, and `erase` members shall affect the validity of iterators and references to the adaptor's container(s) in the same way that the containers' respective `insert`, `emplace`, and `erase` members do.
[Example: A call to `flat_map<Key, T>::insert` invalidates all iterators to the `flat_map`.]
- ⁴ A deduction guide for a container adaptor shall not participate in overload resolution if any of the following are true:
 - (4.1) — It has an `InputIterator` template parameter and a type that does not qualify as an input iterator is deduced for that parameter.
 - (4.2) — It has a `Compare` template parameter and a type that qualifies as an allocator is deduced for that parameter.
 - (4.3) — It has a `Container`, `KeyContainer`, or `MappedContainer` template parameter and a type that qualifies as an allocator is deduced for that parameter.
 - (4.4) — It has an `Allocator` template parameter and a type that does not qualify as an allocator is deduced for that parameter.
 - (4.5) — It has both `Container` and `Allocator` template parameters, and `uses_allocator_v<Container, Allocator>` is false.
 - (4.6) — It has both `KeyContainer` and `Allocator` template parameters, and `uses_allocator_v<KeyContainer, Allocator>` is false.
 - (4.7) — It has both `MappedContainer` and `Allocator` template parameters, and `uses_allocator_v<MappedContainer, Allocator>` is false.

21.6.4 Header `<flat_map>` synopsis

[flatmap.syn]

```
#include <initializer_list>

namespace std {
    // 21.6.8, class template flat_map
    template<class Key, class T, class Compare = less<Key>,
            class KeyContainer = vector<Key>, class MappedContainer = vector<T>>
        class flat_map;

    template<class Key, class T, class Compare,
            class KeyContainer, class MappedContainer>
```

```

    bool operator==(const flat_map<Key, T, Compare, KeyContainer, MappedContainer>& x,
                    const flat_map<Key, T, Compare, KeyContainer, MappedContainer>& y);
template<class Key, class T, class Compare,
        class KeyContainer, class MappedContainer>
    bool operator!=(const flat_map<Key, T, Compare, KeyContainer, MappedContainer>& x,
                    const flat_map<Key, T, Compare, KeyContainer, MappedContainer>& y);
template<class Key, class T, class Compare,
        class KeyContainer, class MappedContainer>
    bool operator< (const flat_map<Key, T, Compare, KeyContainer, MappedContainer>& x,
                    const flat_map<Key, T, Compare, KeyContainer, MappedContainer>& y);
template<class Key, class T, class Compare,
        class KeyContainer, class MappedContainer>
    bool operator> (const flat_map<Key, T, Compare, KeyContainer, MappedContainer>& x,
                    const flat_map<Key, T, Compare, KeyContainer, MappedContainer>& y);
template<class Key, class T, class Compare,
        class KeyContainer, class MappedContainer>
    bool operator<= (const flat_map<Key, T, Compare, KeyContainer, MappedContainer>& x,
                     const flat_map<Key, T, Compare, KeyContainer, MappedContainer>& y);
template<class Key, class T, class Compare,
        class KeyContainer, class MappedContainer>
    bool operator>= (const flat_map<Key, T, Compare, KeyContainer, MappedContainer>& x,
                     const flat_map<Key, T, Compare, KeyContainer, MappedContainer>& y);

template<class Key, class T, class Compare,
        class KeyContainer, class MappedContainer>
    void swap(flat_map<Key, T, Compare, KeyContainer, MappedContainer>& x,
              flat_map<Key, T, Compare, KeyContainer, MappedContainer>& y) noexcept;

struct sorted_unique_t { explicit sorted_unique_t() = default; };
inline constexpr sorted_unique_t sorted_unique {};

// 21.6.9, class template flat_multimap
template<class Key, class T, class Compare = less<Key>,
        class KeyContainer = vector<Key>, class MappedContainer = vector<T>>
    class flat_multimap;

template<class Key, class T, class Compare,
        class KeyContainer, class MappedContainer>
    bool operator==(const flat_multimap<Key, T, Compare, KeyContainer, MappedContainer>& x,
                    const flat_multimap<Key, T, Compare, KeyContainer, MappedContainer>& y);
template<class Key, class T, class Compare,
        class KeyContainer, class MappedContainer>
    bool operator!=(const flat_multimap<Key, T, Compare, KeyContainer, MappedContainer>& x,
                    const flat_multimap<Key, T, Compare, KeyContainer, MappedContainer>& y);
template<class Key, class T, class Compare,
        class KeyContainer, class MappedContainer>
    bool operator< (const flat_multimap<Key, T, Compare, KeyContainer, MappedContainer>& x,
                    const flat_multimap<Key, T, Compare, KeyContainer, MappedContainer>& y);
template<class Key, class T, class Compare,
        class KeyContainer, class MappedContainer>
    bool operator> (const flat_multimap<Key, T, Compare, KeyContainer, MappedContainer>& x,
                    const flat_multimap<Key, T, Compare, KeyContainer, MappedContainer>& y);
template<class Key, class T, class Compare,
        class KeyContainer, class MappedContainer>
    bool operator<= (const flat_multimap<Key, T, Compare, KeyContainer, MappedContainer>& x,

```

```

        const flat_multimap<Key, T, Compare, KeyContainer, MappedContainer>& y);
template<class Key, class T, class Compare,
        class KeyContainer, class MappedContainer>
    bool operator>=(const flat_multimap<Key, T, Compare, KeyContainer, MappedContainer>& x,
        const flat_multimap<Key, T, Compare, KeyContainer, MappedContainer>& y);

template<class Key, class T, class Compare,
        class KeyContainer, class MappedContainer>
    void swap(flat_multimap<Key, T, Compare, KeyContainer, MappedContainer>& x,
        flat_multimap<Key, T, Compare, KeyContainer, MappedContainer>& y) noexcept;

struct sorted_equivalent_t { explicit sorted_equivalent_t() = default; };
inline constexpr sorted_equivalent_t sorted_equivalent {};
}

```

21.6.8 Class template `flat_map`

[flatmap]

- ¹ A `flat_map` is a container adaptor that provides an associative container interface that supports unique keys (contains at most one of each key value) and provides for fast retrieval of values of another type `T` based on the keys. The `flat_map` class supports random access iterators.
- ² A `flat_map` satisfies all of the requirements of a container, of a reversible container (21.2), and of an associative container (21.2.6), except for the requirements related to node handles (21.2.4) and iterator invalidation (21.6.1). A `flat_map` does not meet the additional requirements of an allocator-aware container, as described in Table 80.
- ³ A `flat_map` also provides most operations described in 21.2.6 for unique keys. This means that a `flat_map` supports the `a_uniq` operations in 21.2.6 but not the `a_eq` operations. For a `flat_map<Key,T>` the `key_type` is `Key` and the `value_type` is `pair<const Key,T>`.
- ⁴ A `flat_map` `m` maintains these invariants: it contains the same number of keys and values; the keys are sorted with respect to the comparison object; and the value at offset `o` within the value container is the value associated with the key at offset `o` within the key container.
- ⁵ Descriptions are provided here only for operations on `flat_map` that are not described in one of those tables or for operations where there is additional semantic information.
- ⁶ Any sequence container supporting random access iteration can be used to instantiate `flat_map`. In particular, `vector` (21.3.11) and `deque` (21.3.8) can be used.
- ⁷ The template parameters `Key` and `T` of `flat_map` shall denote the same type as `KeyContainer::value_type` and `MappedContainer::value_type`, respectively.
- ⁸ The effect of calling a constructor that takes both `key_container_type` and `mapped_container_type` arguments with containers of different sizes is undefined.
- ⁹ Constructors that take a `Container` argument `cont` shall participate in overload resolution only if both `std::begin(cont)` and `std::end(cont)` are well-formed expressions.
- ¹⁰ The effect of calling a constructor that takes a `sorted_unique_t` argument with a range that is not sorted with respect to `compare`, or that contains equal elements, is undefined.
- ¹¹ Constructors that take an `Alloc` argument shall participate in overload resolution only if `uses_allocator_v<key_container_type, Alloc> || uses_allocator_v<mapped_container_type, Alloc>` is `true`.
- ¹² Constructors that take an `Alloc` argument shall participate in overload resolution only if `Alloc` meets the allocator requirements as described in (21.2.1).

21.6.8.1 Definition

[flatmap.defn]

```

namespace std {
    template <class Key, class T, class Compare = less<Key>,
              class KeyContainer = vector<Key>,
              class MappedContainer = vector<T>>
    class flat_map {
    public:
        // types:
        using key_type           = Key;
        using mapped_type        = T;
        using value_type          = pair<const key_type, mapped_type>;
        using key_compare         = Compare;
        using reference           = pair<const key_type&, mapped_type&>;
        using const_reference     = pair<const key_type&, const mapped_type&>;
        using size_type           = size_t;
        using difference_type     = ptrdiff_t;
        using iterator            = implementation-defined; // see 21.2
        using const_iterator      = implementation-defined; // see 21.2
        using reverse_iterator    = std::reverse_iterator<iterator>;
        using const_reverse_iterator = std::reverse_iterator<const_iterator>;
        using key_container_type  = KeyContainer;
        using mapped_container_type = MappedContainer;

        class value_compare {
            friend class flat_map;
        protected:
            key_compare comp;
            value_compare(key_compare c) : comp(c) { }
        public:
            bool operator()(const_reference x, const_reference y) const {
                return comp(x.first, y.first);
            }
        };

        struct containers
        {
            key_container_type keys;
            mapped_container_type values;
        };

        // 21.6.8.2, construct/copy/destroy
        flat_map() : flat_map(key_compare()) { }

        flat_map(key_container_type key_cont, mapped_container_type mapped_cont);
        template <class Alloc>
        flat_map(key_container_type key_cont, mapped_container_type mapped_cont,
                 const Alloc& a)
            : flat_map(key_container_type(std::move(key_cont), a),
                      mapped_container_type(std::move(mapped_cont), a))
        { }
        template <class Container>
        explicit flat_map(const Container& cont)
            : flat_map(cont.begin(), cont.end(), key_compare()) { }
        template <class Container, class Alloc>
        flat_map(const Container& cont, const Alloc& a)

```

```

        : flat_map(cont.begin(), cont.end(), key_compare(), a) { }

flat_map(sorted_unique_t,
         key_container_type key_cont, mapped_container_type mapped_cont);
template <class Alloc>
flat_map(sorted_unique_t s, key_container_type key_cont,
         mapped_container_type mapped_cont, const Alloc& a)
    : flat_map(s, key_container_type(std::move(key_cont), a),
              mapped_container_type(std::move(mapped_cont), a))
    { }
template <class Container>
flat_map(sorted_unique_t s, const Container& cont)
    : flat_map(s, cont.begin(), cont.end(), key_compare()) { }
template <class Container, class Alloc>
flat_map(sorted_unique_t s, const Container& cont, const Alloc& a)
    : flat_map(s, cont.begin(), cont.end(), key_compare(), a) { }

explicit flat_map(const key_compare& comp)
    : c(containers()), compare(comp) { }
template <class Alloc>
flat_map(const key_compare& comp, const Alloc& a);
template <class Alloc>
explicit flat_map(const Alloc& a)
    : flat_map(key_compare(), a) { }

template <class InputIterator>
flat_map(InputIterator first, InputIterator last,
         const key_compare& comp = key_compare());
template <class InputIterator, class Alloc>
flat_map(InputIterator first, InputIterator last,
         const key_compare& comp, const Alloc& a);
template <class InputIterator, class Alloc>
flat_map(InputIterator first, InputIterator last,
         const Alloc& a)
    : flat_map(first, last, key_compare(), a) { }

template <class InputIterator>
flat_map(sorted_unique_t, InputIterator first, InputIterator last,
         const key_compare& comp = key_compare());
template <class InputIterator, class Alloc>
flat_map(sorted_unique_t, InputIterator first, InputIterator last,
         const key_compare& comp, const Alloc& a);
template <class InputIterator, class Alloc>
flat_map(sorted_unique_t s, InputIterator first, InputIterator last,
         const Alloc& a)
    : flat_map(s, first, last, key_compare(), a) { }

template <class Alloc>
flat_map(flat_map&& m, const Alloc& a)
    : c{key_container_type(std::move(m.c.keys), a),
      mapped_container_type(std::move(m.c.values), a)},
      compare{std::move(m.compare)}}
    { }
template <class Alloc>
flat_map(const flat_map& m, const Alloc& a)

```



```

        : c{key_container_type(m.c.keys, a),
          mapped_container_type(m.c.values, a)}
        , compare{m.compare}
    { }

flat_map(initializer_list<value_type>&& il,
         const key_compare& comp = key_compare())
    : flat_map(il, comp) { }
template <class Alloc>
flat_map(initializer_list<value_type>&& il,
         const key_compare& comp, const Alloc& a)
    : flat_map(il, comp, a) { }
template <class Alloc>
flat_map(initializer_list<value_type>&& il, const Alloc& a)
    : flat_map(il, key_compare(), a) { }

flat_map(sorted_unique_t s, initializer_list<value_type>&& il,
         const key_compare& comp = key_compare())
    : flat_map(s, il, comp) { }
template <class Alloc>
flat_map(sorted_unique_t s, initializer_list<value_type>&& il,
         const key_compare& comp, const Alloc& a)
    : flat_map(s, il, comp, a) { }
template <class Alloc>
flat_map(sorted_unique_t s, initializer_list<value_type>&& il,
         const Alloc& a)
    : flat_map(s, il, key_compare(), a) { }

flat_map& operator=(initializer_list<value_type> il);

// iterators
iterator          begin() noexcept;
const_iterator    begin() const noexcept;
iterator          end() noexcept;
const_iterator    end() const noexcept;

reverse_iterator  rbegin() noexcept;
const_reverse_iterator rbegin() const noexcept;
reverse_iterator  rend() noexcept;
const_reverse_iterator rend() const noexcept;

const_iterator    cbegin() const noexcept;
const_iterator    cend() const noexcept;
const_reverse_iterator crbegin() const noexcept;
const_reverse_iterator crend() const noexcept;

// 21.6.8.4, capacity
[[nodiscard]] bool empty() const noexcept;
size_type size() const noexcept;
size_type max_size() const noexcept;

// 21.6.8.5, element access
mapped_type& operator[](const key_type& x);
mapped_type& operator[](key_type&& x);
mapped_type& at(const key_type& x);

```

```

const mapped_type& at(const key_type& x) const;

// 21.6.8.6, modifiers
template <class... Args> pair<iterator, bool> emplace(Args&&... args);
template <class... Args>
    iterator emplace_hint(const_iterator position, Args&&... args);
pair<iterator, bool> insert(const value_type& x);
pair<iterator, bool> insert(value_type&& x);
template <class P> pair<iterator, bool> insert(P&& x);
iterator insert(const_iterator position, const value_type& x);
iterator insert(const_iterator position, value_type&& x);
template <class P>
    iterator insert(const_iterator position, P&&);
template <class InputIterator>
    void insert(InputIterator first, InputIterator last);
template <class InputIterator>
    void insert(sorted_unique_t, InputIterator first, InputIterator last);
void insert(initializer_list<value_type>);
void insert(sorted_unique_t, initializer_list<value_type> il);

containers extract() &&;
void replace(key_container_type&& key_cont, mapped_container_type&& mapped_cont);

template <class... Args>
    pair<iterator, bool> try_emplace(const key_type& k, Args&&... args);
template <class... Args>
    pair<iterator, bool> try_emplace(key_type&& k, Args&&... args);
template <class... Args>
    iterator try_emplace(const_iterator hint, const key_type& k,
        Args&&... args);
template <class... Args>
    iterator try_emplace(const_iterator hint, key_type&& k, Args&&... args);
template <class M>
    pair<iterator, bool> insert_or_assign(const key_type& k, M&& obj);
template <class M>
    pair<iterator, bool> insert_or_assign(key_type&& k, M&& obj);
template <class M>
    iterator insert_or_assign(const_iterator hint, const key_type& k,
        M&& obj);
template <class M>
    iterator insert_or_assign(const_iterator hint, key_type&& k, M&& obj);

iterator erase(iterator position);
iterator erase(const_iterator position);
size_type erase(const key_type& x);
iterator erase(const_iterator first, const_iterator last);

void swap(flat_map& fm) noexcept;
void clear() noexcept;

// observers
key_compare key_comp() const;
value_compare value_comp() const;

const key_container_type& keys() const    { return c.keys; }

```

```

const mapped_container_type& values() const { return c.values; }

// map operations
iterator find(const key_type& x);
const_iterator find(const key_type& x) const;
template <class K> iterator find(const K& x);
template <class K> const_iterator find(const K& x) const;

size_type count(const key_type& x) const;
template <class K> size_type count(const K& x) const;

bool contains(const key_type& x) const;
template <class K> bool contains(const K& x) const;

iterator lower_bound(const key_type& x);
const_iterator lower_bound(const key_type& x) const;
template <class K> iterator lower_bound(const K& x);
template <class K> const_iterator lower_bound(const K& x) const;

iterator upper_bound(const key_type& x);
const_iterator upper_bound(const key_type& x) const;
template <class K> iterator upper_bound(const K& x);
template <class K> const_iterator upper_bound(const K& x) const;

pair<iterator, iterator> equal_range(const key_type& x);
pair<const_iterator, const_iterator> equal_range(const key_type& x) const;
template <class K>
    pair<iterator, iterator> equal_range(const K& x);
template <class K>
    pair<const_iterator, const_iterator> equal_range(const K& x) const;

private:
    containers c;           // exposition only
    key_compare compare;    // exposition only
};

template<class Container>
    using cont-key-type =
        remove_const_t<typename Container::value_type::first_type>;           // exposition only
template<class Container>
    using cont-mapped-type =
        typename Container::value_type::second_type;           // exposition only
template<class InputIterator>
    using iter-key-type = remove_const_t<
        typename iterator_traits<InputIterator>::value_type::first_type>;    // exposition only
template<class InputIterator>
    using iter-mapped-type =
        typename iterator_traits<InputIterator>::value_type::second_type;    // exposition only

template <class Container>
    flat_map(Container)
        -> flat_map<cont-key-type <Container>, cont-mapped-type <Container>>;

template <class KeyContainer, class MappedContainer>
    flat_map(KeyContainer, MappedContainer)

```

```

    -> flat_map<typename KeyContainer::value_type,
                typename MappedContainer::value_type,
                less<typename KeyContainer::value_type>,
                KeyContainer, MappedContainer>;

template <class Container, class Alloc>
flat_map(Container, Alloc)
    -> flat_map<cont-key-type <Container>, cont-mapped-type <Container>>;

template <class KeyContainer, class MappedContainer, class Alloc>
flat_map(KeyContainer, MappedContainer, Alloc)
    -> flat_map<typename KeyContainer::value_type,
                typename MappedContainer::value_type,
                less<typename KeyContainer::value_type>,
                KeyContainer, MappedContainer>;

template <class Container>
flat_map(sorted_unique_t, Container)
    -> flat_map<cont-key-type <Container>, cont-mapped-type <Container>>;

template <class KeyContainer, class MappedContainer>
flat_map(sorted_unique_t, KeyContainer, MappedContainer)
    -> flat_map<typename KeyContainer::value_type,
                typename MappedContainer::value_type,
                less<typename KeyContainer::value_type>,
                KeyContainer, MappedContainer>;

template <class Container, class Alloc>
flat_map(sorted_unique_t, Container, Alloc)
    -> flat_map<cont-key-type <Container>, cont-mapped-type <Container>>;

template <class KeyContainer, class MappedContainer, class Alloc>
flat_map(sorted_unique_t, KeyContainer, MappedContainer, Alloc)
    -> flat_map<typename KeyContainer::value_type,
                typename MappedContainer::value_type,
                less<typename KeyContainer::value_type>,
                KeyContainer, MappedContainer>;

template <class InputIterator, class Compare = less<iter-mapped-type <InputIterator>>>
flat_map(InputIterator, InputIterator, Compare = Compare())
    -> flat_map<iter-mapped-type <InputIterator>, iter-mapped-type <InputIterator>, Compare>;

template<class InputIterator, class Compare, class Alloc>
flat_map(InputIterator, InputIterator, Compare, Alloc)
    -> flat_map<iter-mapped-type <InputIterator>, iter-mapped-type <InputIterator>, Compare>;

template<class InputIterator, class Alloc>
flat_map(InputIterator, InputIterator, Alloc)
    -> flat_map<iter-mapped-type <InputIterator>, iter-mapped-type <InputIterator>>;

template <class InputIterator, class Compare = less<iter-mapped-type <InputIterator>>>
flat_map(sorted_unique_t, InputIterator, InputIterator, Compare = Compare())
    -> flat_map<iter-mapped-type <InputIterator>, iter-mapped-type <InputIterator>, Compare>;

template<class InputIterator, class Compare, class Alloc>

```

```

    flat_map(sorted_unique_t, InputIterator, InputIterator, Compare, Alloc)
    -> flat_map<iter-mapped-type <InputIterator>, iter-mapped-type <InputIterator>, Compare>;

template<class InputIterator, class Alloc>
    flat_map(sorted_unique_t, InputIterator, InputIterator, Alloc)
    -> flat_map<iter-mapped-type <InputIterator>, iter-mapped-type <InputIterator>>;

template<class Key, class T, class Compare = less<Key>>
    flat_map(initializer_list<pair<const Key, T>>, Compare = Compare())
    -> flat_map<Key, T, Compare>;

template<class Key, class T, class Compare, class Alloc>
    flat_map(initializer_list<pair<const Key, T>>, Compare, Alloc)
    -> flat_map<Key, T, Compare>;

template<class Key, class T, class Alloc>
    flat_map(initializer_list<pair<const Key, T>>, Alloc)
    -> flat_map<Key, T>;

template<class Key, class T, class Compare = less<Key>>
    flat_map(sorted_unique_t, initializer_list<pair<const Key, T>>, Compare = Compare())
    -> flat_map<Key, T, Compare>;

template<class Key, class T, class Compare, class Alloc>
    flat_map(sorted_unique_t, initializer_list<pair<const Key, T>>, Compare, Alloc)
    -> flat_map<Key, T, Compare>;

template<class Key, class T, class Alloc>
    flat_map(sorted_unique_t, initializer_list<pair<const Key, T>>, Alloc)
    -> flat_map<Key, T>;
}

```

21.6.8.2 Constructors

[flatmap.cons]

```
flat_map(key_container_type key_cont, mapped_container_type mapped_cont);
```

- 1 *Effects:* Initializes `c.keys` with `std::move(key_cont)` and `c.values` with `std::move(mapped_cont)`; value-initializes `compare`; and sorts the range `[begin(),end())` with `compare`.
- 2 *Complexity:* Linear in N if the container arguments are already sorted as if with `compare` and otherwise $N \log N$, where N is `key_cont.size()`.

```
flat_map(sorted_unique_t, key_container_type key_cont, mapped_container_type mapped_cont);
```

- 3 *Effects:* Initializes `c.keys` with `std::move(key_cont)` and `c.values` with `std::move(mapped_cont)`; value-initializes `compare`.
- 4 *Complexity:* Constant.

```

template <class InputIterator>
    flat_map(InputIterator first, InputIterator last, const key_compare& comp);

```

- 5 *Effects:* Initializes `compare` with `comp`, adds elements to `c` as if by:

```

        for (; first != last; ++first) {
            c.keys.insert(c.keys.end(), first->first);
            c.values.insert(c.values.end(), first->second);
        }

```

and finally sorts the range `[begin(),end())` with `compare`.

- 6 *Complexity:* Linear in N if the container arguments are already sorted as if with `compare` and otherwise $N \log N$, where N is `key_cont.size()`.

```
template <class InputIterator>
flat_map(sorted_unique_t, InputIterator first, InputIterator last,
         const key_compare& comp = key_compare());
```

- 7 *Effects:* Initializes `compare` with `comp`, and adds elements to `c.keys` and `c.values` as if by:

```
for (; first != last; ++first) {
    c.keys.insert(c.keys.end(), first->first);
    c.values.insert(c.values.end(), first->second);
}
```

- 8 *Complexity:* Linear.

21.6.8.3 Constructors with allocators

[flatmap.cons.alloc]

```
template <class Alloc>
flat_map(const key_compare& comp, const Alloc& a);
```

- 1 *Effects:* Initializes `compare` with `comp`, and performs uses-allocator construction (23.10.8.2) of both `c.keys` and `c.values` with `a`.

- 2 *Complexity:* Constant.

```
template <class InputIterator, class Alloc>
flat_map(InputIterator first, InputIterator last,
         const key_compare& comp, const Alloc& a);
```

- 3 *Effects:* Initializes `compare` with `comp`, and performs uses-allocator construction (23.10.8.2) of both `c.keys` and `c.values` with `a`; adds elements to `c.keys` and `c.values` as if by:

```
for (; first != last; ++first) {
    c.keys.insert(c.keys.end(), first->first);
    c.values.insert(c.values.end(), first->second);
}
```

and finally sorts the range `[begin(),end())` with `compare`.

- 4 *Complexity:* Linear in N if the container arguments are already sorted as if with `compare` and otherwise $N \log N$, where N is `distance(first, last)`.

```
template <class InputIterator, class Alloc>
flat_map(sorted_unique_t, InputIterator first, InputIterator last,
         const key_compare& comp, const Alloc& a);
```

- 5 *Effects:* Initializes `compare` with `comp`, and performs uses-allocator construction (23.10.8.2) of both `c.keys` and `c.values` with `a`; adds elements to `c.keys` and `c.values` as if by:

```
for (; first != last; ++first) {
    c.keys.insert(c.keys.end(), first->first);
    c.values.insert(c.values.end(), first->second);
}
```

- 6 *Complexity:* Linear.

21.6.8.4 Capacity**[flatmap.capacity]**

```
size_type size() const noexcept;
```

1 *Effects:* Equivalent to: `return c.keys.size();`

```
size_type max_size() const noexcept;
```

2 *Effects:* Equivalent to: `return std::min<size_type>(c.keys.max_size(), c.values.max_size());`

21.6.8.5 Access**[flatmap.access]**

```
mapped_type& operator[](const key_type& x);
```

1 *Effects:* Equivalent to: `return try_emplace(x).first->second;`

```
mapped_type& operator[](key_type&& x);
```

2 *Effects:* Equivalent to: `return try_emplace(std::move(x)).first->second;`

```
mapped_type& at(const key_type& x);
```

```
const mapped_type& at(const key_type& x) const;
```

3 *Returns:* A reference to the `mapped_type` corresponding to `x` in `*this`.

4 *Throws:* An exception object of type `out_of_range` if no such element is present.

5 *Complexity:* Logarithmic.

21.6.8.6 Modifiers**[flatmap.modifiers]**

```
flat_map& operator=(initializer_list<value_type> il);
```

1 *Effects:* Equivalent to:

```
clear();
insert(il);
```

```
template <class... Args> pair<iterator, bool> emplace(Args&&... args);
```

2 *Constraints:* `pair<key_type, mapped_type>(std::forward<Args>(args)...) is well-formed.`

3 *Effects:* First, constructs a `pair<key_type, value_type>` object `t` constructed with `std::forward<Args>(args)...`. If the map already contains an element whose key is equivalent to the key of `t`, there is no effect. Otherwise, equivalent to:

```
auto key_it = lower_bound(c.keys.begin(), c.keys.end(), t.first);
auto value_it = c.values.begin() + distance(c.keys.begin(), key_it);
c.keys.emplace(key_it, std::move(t.first));
c.values.emplace(value_it, std::move(t.second));
```

4 *Returns:* The `bool` component of the returned pair is `true` if and only if the insertion took place, and the iterator component of the pair points to the element with key equivalent to the key of `t`.

```
template<class P> pair<iterator, bool> insert(P&& x);
```

```
template<class P> iterator insert(const_iterator position, P&& x);
```

5 *Constraints:* `pair<key_type, mapped_type>(std::forward<P>(x)) is well-formed.`

6 *Effects:* The first form is equivalent to `return emplace(std::forward<P>(x))`. The second form is equivalent to `return emplace_hint(position, std::forward<P>(x))`.

```

template<class... Args>
    pair<iterator, bool> try_emplace(const key_type& k, Args&&... args);
template<class... Args>
    iterator try_emplace(const_iterator hint, const key_type& k, Args&&... args);

```

7 *Constraints:* mapped_type(std::forward<Args>(args)...) is well-formed.

8 *Effects:* If the map already contains an element whose key is equivalent to k, there is no effect. Otherwise equivalent to:

```

        auto key_it = lower_bound(c.keys.begin(), c.keys.end(), k);
        auto value_it = c.values.begin() + distance(c.keys.begin(), key_it);
        c.keys.insert(key_it, k);
        c.values.emplace(value_it, std::forward<Args>(args)...);

```

9 *Returns:* In the first overload, the bool component of the returned pair is true if and only if the insertion took place. The returned iterator points to the map element whose key is equivalent to k.

10 *Complexity:* The same as emplace and emplace_hint, respectively.

```

template<class... Args>
    pair<iterator, bool> try_emplace(key_type&& k, Args&&... args);
template<class... Args>
    iterator try_emplace(const_iterator hint, key_type&& k, Args&&... args);

```

11 *Constraints:* mapped_type(std::forward<Args>(args)...) is well-formed.

12 *Effects:* If the map already contains an element whose key is equivalent to k, there is no effect. Otherwise equivalent to:

```

        auto key_it = lower_bound(c.keys.begin(), c.keys.end(), k);
        auto value_it = c.values.begin() + distance(c.keys.begin(), key_it);
        c.keys.emplace(key_it, std::move(k));
        c.values.emplace(value_it, std::forward<Args>(args)...);

```

13 *Returns:* In the first overload, the bool component of the returned pair is true if and only if the insertion took place. The returned iterator points to the map element whose key is equivalent to k.

14 *Complexity:* The same as emplace and emplace_hint, respectively.

```

template<class M>
    pair<iterator, bool> insert_or_assign(const key_type& k, M&& obj);
template<class M>
    iterator insert_or_assign(const_iterator hint, const key_type& k, M&& obj);

```

15 *Constraints:* is_assignable_v<mapped_type&, M> is true, and mapped_type(std::forward<M>(obj)) is well-formed.

16 *Effects:* If the map already contains an element e whose key is equivalent to k, assigns std::forward<M>(obj) to e.second. Otherwise equivalent to insert(k, std::forward<M>(obj)) or emplace(hint, k, std::forward<M>(obj)) respectively.

17 *Returns:* In the first overload, the bool component of the returned pair is true if and only if the insertion took place. The returned iterator points to the map element whose key is equivalent to k.

18 *Complexity:* The same as emplace and emplace_hint, respectively.

```

template<class M>
    pair<iterator, bool> insert_or_assign(key_type&& k, M&& obj);
template<class M>
    iterator insert_or_assign(const_iterator hint, key_type&& k, M&& obj);

```


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Constraints: `is_assignable_v<mapped_type&, M>` is true, and the expression `mapped_type(std::forward<M>(obj))` is well-formed.

Effects: If the map already contains an element `e` whose key is equivalent to `k`, assigns `std::forward<M>(obj)` to `e.second`. Otherwise equivalent to `insert(std::move(k), std::forward<M>(obj))` or `emplace(hint, std::move(k), std::forward<M>(obj))` respectively.

Returns: In the first overload, the `bool` component of the returned pair is `true` if and only if the insertion took place. The returned iterator points to the map element whose key is equivalent to `k`.

Complexity: The same as `emplace` and `emplace_hint`, respectively.

```
template <class InputIterator>
    void insert(sorted_unique_t, InputIterator first, InputIterator last);
```

Expects: The range `[first,last)` is sorted with respect to `compare`.

Effects: Equivalent to: `insert(first, last)`.

Complexity: Linear.

```
void insert(sorted_unique_t, initializer_list<value_type> il);
```

Effects: Equivalent to `insert(sorted_unique_t, il.begin(), il.end())`.

```
void swap(flat_map& fm) noexcept;
```

Constraints: `is_nothrow_swappable_v<key_container_type> && is_nothrow_swappable_v<mapped_container_type> && is_nothrow_swappable_v<key_compare>` is true.

Effects: Equivalent to:

```
    using std::swap;
    swap(c.keys, fm.c.keys);
    swap(c.values, fm.c.values);
    swap(c.compare, fm.compare);
```

```
containers extract() &&;
```

Effects: Equivalent to:

```
    containers temp;
    temp.keys.swap(c.keys);
    temp.values.swap(c.values);
    return temp;
```

```
void replace(key_container_type&& key_cont, mapped_container_type&& mapped_cont);
```

Expects: `key_cont.size() == mapped_cont.size()` is true, and the elements of `key_cont` are sorted with respect to `compare`.

Effects: Equivalent to:

```
    c.keys = std::move(key_cont);
    c.values = std::move(mapped_cont);
```

21.6.8.7 Operators**[flatmap.ops]**

```
template<class Key, class T, class Compare, class KeyContainer, class MappedContainer>
    bool operator==(const flat_map<Key, T, Compare, KeyContainer, MappedContainer>& x,
                    const flat_map<Key, T, Compare, KeyContainer, MappedContainer>& y);
```

¹ *Effects:* Equivalent to: `return std::equal(x.begin(), x.end(), y.begin(), y.end());`

```
template<class Key, class T, class Compare, class KeyContainer, class MappedContainer>
    bool operator!=(const flat_map<Key, T, Compare, KeyContainer, MappedContainer>& x,
                    const flat_map<Key, T, Compare, KeyContainer, MappedContainer>& y);
```

² *Returns:* `!(x == y)`.

```
template<class Key, class T, class Compare, class KeyContainer, class MappedContainer>
    bool operator< (const flat_map<Key, T, Compare, KeyContainer, MappedContainer>& x,
                    const flat_map<Key, T, Compare, KeyContainer, MappedContainer>& y);
```

³ *Effects:* Equivalent to: `return std::lexicographical_compare(x.begin(), x.end(), y.begin(), y.end());`

```
template<class Key, class T, class Compare, class KeyContainer, class MappedContainer>
    bool operator> (const flat_map<Key, T, Compare, KeyContainer, MappedContainer>& x,
                    const flat_map<Key, T, Compare, KeyContainer, MappedContainer>& y);
```

⁴ *Returns:* `y < x`.

```
template<class Key, class T, class Compare, class KeyContainer, class MappedContainer>
    bool operator<=(const flat_map<Key, T, Compare, KeyContainer, MappedContainer>& x,
                    const flat_map<Key, T, Compare, KeyContainer, MappedContainer>& y);
```

⁵ *Returns:* `!(y < x)`.

```
template<class Key, class T, class Compare, class KeyContainer, class MappedContainer>
    bool operator>=(const flat_map<Key, T, Compare, KeyContainer, MappedContainer>& x,
                    const flat_map<Key, T, Compare, KeyContainer, MappedContainer>& y);
```

⁶ *Returns:* `!(x < y)`.

21.6.8.8 Specialized algorithms**[flatmap.special]**

```
template<class Key, class T, class Compare, class KeyContainer, class MappedContainer>
    void swap(flat_map<Key, T, Compare, KeyContainer, MappedContainer>& x,
              flat_map<Key, T, Compare, KeyContainer, MappedContainer>& y) noexcept;
```

¹ *Constraints:* `is_nothrow_swappable_v<KeyContainer> && is_nothrow_swappable_v<MappedContainer> && is_nothrow_swappable_v<Compare>` is true.

² *Effects:* Equivalent to: `x.swap(y)`.

21.6.9 Class template flat_multimap**[flatmultimap]**

¹ A `flat_multimap` is a container adaptor that provides an associative container interface that supports equivalent keys (possibly containing multiple copies of the same key value) and provides for fast retrieval of values of another type `T` based on the keys. The `flat_multimap` class supports random access iterators.

² A `flat_multimap` satisfies all of the requirements of a container, of a reversible container (21.2), and of an associative container (21.2.6), except for the requirements related to node handles (21.2.4) and iterator invalidation (21.6.1). A `flat_multimap` does not meet the additional requirements of an allocator-aware container, as described in Table 80.

- ³ A `flat_multimap` also provides most operations described in 21.2.6 for equal keys. This means that a `flat_multimap` supports the `a_eq` operations in 21.2.6 but not the `a_uniq` operations. For a `flat_multimap<Key,T>` the `key_type` is `Key` and the `value_type` is `pair<const Key,T>`.
- ⁴ A `flat_multimap` `m` maintains these invariants: it contains the same number of keys and values; the keys are sorted with respect to the comparison object; and the value at offset `o` within the value container is the value associated with the key at offset `o` within the key container.
- ⁵ Descriptions are provided here only for operations on `flat_multimap` that are not described in one of those tables or for operations where there is additional semantic information.
- ⁶ Any sequence container supporting random access iteration can be used to instantiate `flat_multimap`. In particular, `vector` (21.3.11) and `deque` (21.3.8) can be used.
- ⁷ The template parameters `Key` and `T` of `flat_multimap` shall denote the same type as `KeyContainer::value_type` and `MappedContainer::value_type`, respectively.
- ⁸ The effect of calling a constructor that takes both `key_container_type` and `mapped_container_type` arguments with containers of different sizes is undefined.
- ⁹ Constructors that take a `Container` argument `cont` shall participate in overload resolution only if both `std::begin(cont)` and `std::end(cont)` are well-formed expressions.
- ¹⁰ The effect of calling a constructor that takes a `sorted_equivalent_t` argument with a container or containers that are not sorted with respect to `key_compare` is undefined.
- ¹¹ Constructors that take an `Alloc` argument shall participate in overload resolution only if `uses_allocator_v<key_container_type, Alloc> || uses_allocator_v<mapped_container_type, Alloc>` is true.
- ¹² Constructors that take an `Alloc` argument shall participate in overload resolution only if `Alloc` meets the allocator requirements as described in (21.2.1).

21.6.9.1 Definition

[flatmultimap.defn]

```
namespace std {
    template <class Key, class T, class Compare = less<Key>,
              class KeyContainer = vector<Key>,
              class MappedContainer = vector<T>>
    class flat_multimap {
    public:
        // types:
        using key_type           = Key;
        using mapped_type        = T;
        using value_type         = pair<const key_type, mapped_type>;
        using key_compare        = Compare;
        using reference           = pair<const key_type&, mapped_type&>;
        using const_reference    = pair<const key_type&, const mapped_type&>;
        using size_type          = size_t;
        using difference_type     = ptrdiff_t;
        using iterator           = implementation-defined; // see 21.2
        using const_iterator     = implementation-defined; // see 21.2
        using reverse_iterator    = std::reverse_iterator<iterator>;
        using const_reverse_iterator = std::reverse_iterator<const_iterator>;
        using key_container_type  = KeyContainer;
        using mapped_container_type = MappedContainer;

        class value_compare {
            friend class flat_multimap;
        protected:
            key_compare comp;
```

```

    value_compare(key_compare c) : comp(c) { }
public:
    bool operator()(const_reference x, const_reference y) const {
        return comp(x.first, y.first);
    }
};

struct containers
{
    key_container_type keys;
    mapped_container_type values;
};

// 21.6.9.2, construct/copy/destroy
flat_multimap() : flat_multimap(key_compare()) { }

flat_multimap(key_container_type key_cont, mapped_container_type mapped_cont);
template <class Alloc>
flat_multimap(key_container_type key_cont, mapped_container_type mapped_cont,
               const Alloc& a)
    : flat_multimap(key_container_type(std::move(key_cont), a),
                    mapped_container_type(std::move(mapped_cont), a))
{ }
template <class Container>
explicit flat_multimap(const Container& cont)
    : flat_multimap(cont.begin(), cont.end(), key_compare()) { }
template <class Container, class Alloc>
flat_multimap(const Container& cont, const Alloc& a)
    : flat_multimap(cont.begin(), cont.end(), key_compare(), a) { }

flat_multimap(sorted_equivalent_t,
               key_container_type key_cont, mapped_container_type mapped_cont);
template <class Alloc>
flat_multimap(sorted_equivalent_t, key_container_type key_cont,
               mapped_container_type mapped_cont, const Alloc& a)
    : flat_multimap(key_container_type(std::move(key_cont), a),
                    mapped_container_type(std::move(mapped_cont), a))
{ }
template <class Container>
flat_multimap(sorted_equivalent_t s, const Container& cont)
    : flat_multimap(s, cont.begin(), cont.end(), key_compare()) { }
template <class Container, class Alloc>
flat_multimap(sorted_equivalent_t s, const Container& cont, const Alloc& a)
    : flat_multimap(s, cont.begin(), cont.end(), key_compare(), a) { }

explicit flat_multimap(const key_compare& comp)
    : c(containers()), compare(comp) { }
template <class Alloc>
flat_multimap(const key_compare& comp, const Alloc& a);
template <class Alloc>
explicit flat_multimap(const Alloc& a)
    : flat_multimap(key_compare(), a) { }

template <class InputIterator>
flat_multimap(InputIterator first, InputIterator last,

```

```

        const key_compare& comp = key_compare();
template <class InputIterator, class Alloc>
    flat_multimap(InputIterator first, InputIterator last,
        const key_compare& comp, const Alloc& a);
template <class InputIterator, class Alloc>
    flat_multimap(InputIterator first, InputIterator last,
        const Alloc& a)
        : flat_multimap(first, last, key_compare(), a) { }

template <class InputIterator>
    flat_multimap(sorted_equivalent_t, InputIterator first, InputIterator last,
        const key_compare& comp = key_compare());
template <class InputIterator, class Alloc>
    flat_multimap(sorted_equivalent_t, InputIterator first, InputIterator last,
        const key_compare& comp, const Alloc& a);
template <class InputIterator, class Alloc>
    flat_multimap(sorted_equivalent_t s, InputIterator first, InputIterator last,
        const Alloc& a)
        : flat_multimap(s, first, last, key_compare(), a) { }

template <class Alloc>
    flat_multimap(flat_multimap&& m, const Alloc& a)
        : c{key_container_type(std::move(m.c.keys), a),
            mapped_container_type(std::move(m.c.values), a)}
            , compare{std::move(m.compare)}
        { }
template<class Alloc>
    flat_multimap(const flat_multimap& m, const Alloc& a)
        : c{key_container_type(m.c.keys, a), mapped_container_type(m.c.values, a)}
            , compare{m.compare}
        { }

flat_multimap(initializer_list<value_type>&& il,
    const key_compare& comp = key_compare())
    : flat_multimap(il, comp) { }
template <class Alloc>
    flat_multimap(initializer_list<value_type>&& il,
        const key_compare& comp, const Alloc& a)
        : flat_multimap(il, comp, a) { }
template <class Alloc>
    flat_multimap(initializer_list<value_type>&& il, const Alloc& a)
        : flat_multimap(il, key_compare(), a) { }

flat_multimap(sorted_equivalent_t s, initializer_list<value_type>&& il,
    const key_compare& comp = key_compare())
    : flat_multimap(s, il, comp) { }
template <class Alloc>
    flat_multimap(sorted_equivalent_t s, initializer_list<value_type>&& il,
        const key_compare& comp, const Alloc& a)
        : flat_multimap(s, il, comp, a) { }
template <class Alloc>
    flat_multimap(sorted_equivalent_t s, initializer_list<value_type>&& il,
        const Alloc& a)
        : flat_multimap(s, il, key_compare(), a) { }

```

```

flat_multimap& operator=(initializer_list<value_type> il);

// iterators
iterator          begin() noexcept;
const_iterator    begin() const noexcept;
iterator          end() noexcept;
const_iterator    end() const noexcept;

reverse_iterator  rbegin() noexcept;
const_reverse_iterator rbegin() const noexcept;
reverse_iterator  rend() noexcept;
const_reverse_iterator rend() const noexcept;

const_iterator    cbegin() const noexcept;
const_iterator    cend() const noexcept;
const_reverse_iterator crbegin() const noexcept;
const_reverse_iterator crend() const noexcept;

// capacity
[[nodiscard]] bool empty() const noexcept;
size_type size() const noexcept;
size_type max_size() const noexcept;

// 21.6.9.5, modifiers
template <class... Args> pair<iterator, bool> emplace(Args&&... args);
template <class... Args>
    iterator emplace_hint(const_iterator position, Args&&... args);
pair<iterator, bool> insert(const value_type& x);
pair<iterator, bool> insert(value_type&& x);
template <class P> pair<iterator, bool> insert(P&& x);
iterator insert(const_iterator position, const value_type& x);
iterator insert(const_iterator position, value_type&& x);
template <class P>
    iterator insert(const_iterator position, P&&);
template <class InputIterator>
    void insert(InputIterator first, InputIterator last);
template <class InputIterator>
    void insert(sorted_equivalent_t, InputIterator first, InputIterator last);
void insert(initializer_list<value_type>);
void insert(sorted_equivalent_t, initializer_list<value_type> il);

containers extract() &&;
void replace(key_container_type&& key_cont, mapped_container_type&& mapped_cont);

iterator erase(iterator position);
iterator erase(const_iterator position);
size_type erase(const key_type& x);
iterator erase(const_iterator first, const_iterator last);

void swap(flat_multimap& fm) noexcept;
void clear() noexcept;

// observers
key_compare key_comp() const;
value_compare value_comp() const;

```

```

const key_container_type& keys() const      { return c.keys; }
const mapped_container_type& values() const { return c.values; }

// map operations
iterator find(const key_type& x);
const_iterator find(const key_type& x) const;
template <class K> iterator find(const K& x);
template <class K> const_iterator find(const K& x) const;

size_type count(const key_type& x) const;
template <class K> size_type count(const K& x) const;

bool contains(const key_type& x) const;
template <class K> bool contains(const K& x) const;

iterator lower_bound(const key_type& x);
const_iterator lower_bound(const key_type& x) const;
template <class K> iterator lower_bound(const K& x);
template <class K> const_iterator lower_bound(const K& x) const;

iterator upper_bound(const key_type& x);
const_iterator upper_bound(const key_type& x) const;
template <class K> iterator upper_bound(const K& x);
template <class K> const_iterator upper_bound(const K& x) const;

pair<iterator, iterator> equal_range(const key_type& x);
pair<const_iterator, const_iterator> equal_range(const key_type& x) const;
template <class K>
    pair<iterator, iterator> equal_range(const K& x);
template <class K>
    pair<const_iterator, const_iterator> equal_range(const K& x) const;

private:
    containers c;           // exposition only
    key_compare compare;    // exposition only
};

template<class Container>
    using cont-key-type =
        remove_const_t<typename Container::value_type::first_type>;           // exposition only
template<class Container>
    using cont-mapped-type =
        typename Container::value_type::second_type;           // exposition only
template<class InputIterator>
    using iter-key-type = remove_const_t<
        typename iterator_traits<InputIterator>::value_type::first_type>;    // exposition only
template<class InputIterator>
    using iter-mapped-type =
        typename iterator_traits<InputIterator>::value_type::second_type;    // exposition only

template <class Container>
    flat_multimap(Container)
        -> flat_multimap<cont-key-type <Container>, cont-mapped-type <Container>>;

```

```

template <class KeyContainer, class MappedContainer>
flat_multimap(KeyContainer, MappedContainer)
-> flat_multimap<typename KeyContainer::value_type,
                typename MappedContainer::value_type,
                less<typename KeyContainer::value_type>,
                KeyContainer, MappedContainer>;

template <class Container, class Alloc>
flat_multimap(Container, Alloc)
-> flat_multimap<cont-key-type <Container>, cont-mapped-type <Container>>;

template <class KeyContainer, class MappedContainer, class Alloc>
flat_multimap(KeyContainer, MappedContainer, Alloc)
-> flat_multimap<typename KeyContainer::value_type,
                typename MappedContainer::value_type,
                less<typename KeyContainer::value_type>,
                KeyContainer, MappedContainer>;

template <class Container>
flat_multimap(sorted_equivalent_t, Container)
-> flat_multimap<cont-key-type <Container>, cont-mapped-type <Container>>;

template <class KeyContainer, class MappedContainer>
flat_multimap(sorted_equivalent_t, KeyContainer, MappedContainer)
-> flat_multimap<typename KeyContainer::value_type,
                typename MappedContainer::value_type,
                less<typename KeyContainer::value_type>,
                KeyContainer, MappedContainer>;

template <class Container, class Alloc>
flat_multimap(sorted_equivalent_t, Container, Alloc)
-> flat_multimap<cont-key-type <Container>, cont-mapped-type <Container>>;

template <class KeyContainer, class MappedContainer, class Alloc>
flat_multimap(sorted_equivalent_t, KeyContainer, MappedContainer, Alloc)
-> flat_multimap<typename KeyContainer::value_type,
                typename MappedContainer::value_type,
                less<typename KeyContainer::value_type>,
                KeyContainer, MappedContainer>;

template <class InputIterator, class Compare = less<iter-key-type <InputIterator>>>
flat_multimap(InputIterator, InputIterator, Compare = Compare())
-> flat_multimap<iter-key-type <InputIterator>, iter-mapped-type <InputIterator>, Compare>;

template<class InputIterator, class Compare, class Alloc>
flat_multimap(InputIterator, InputIterator, Compare, Alloc)
-> flat_multimap<iter-key-type <InputIterator>, iter-mapped-type <InputIterator>, Compare>;

template<class InputIterator, class Alloc>
flat_multimap(InputIterator, InputIterator, Alloc)
-> flat_multimap<iter-key-type <InputIterator>, iter-mapped-type <InputIterator>>;

template <class InputIterator, class Compare = less<iter-key-type <InputIterator>>>
flat_multimap(sorted_equivalent_t, InputIterator, InputIterator,
              Compare = Compare())

```



```

-> flat_multimap<iter-key-type <InputIterator>, iter-mapped-type <InputIterator>, Compare>;

template<class InputIterator, class Compare, class Alloc>
flat_multimap(sorted_equivalent_t, InputIterator, InputIterator, Compare, Alloc)
-> flat_multimap<iter-key-type <InputIterator>, iter-mapped-type <InputIterator>, Compare>;

template<class InputIterator, class Alloc>
flat_multimap(sorted_equivalent_t, InputIterator, InputIterator, Alloc)
-> flat_multimap<iter-key-type <InputIterator>, iter-mapped-type <InputIterator>>;

template<class Key, class T, class Compare = less<Key>>
flat_multimap(initializer_list<pair<const Key, T>>, Compare = Compare())
-> flat_multimap<Key, T, Compare>;

template<class Key, class T, class Compare, class Alloc>
flat_multimap(initializer_list<pair<const Key, T>>, Compare, Alloc)
-> flat_multimap<Key, T, Compare>;

template<class Key, class T, class Alloc>
flat_multimap(initializer_list<pair<const Key, T>>, Alloc)
-> flat_multimap<Key, T>;

template<class Key, class T, class Compare = less<Key>>
flat_multimap(sorted_equivalent_t, initializer_list<pair<const Key, T>>,
               Compare = Compare())
-> flat_multimap<Key, T, Compare>;

template<class Key, class T, class Compare, class Alloc>
flat_multimap(sorted_equivalent_t, initializer_list<pair<const Key, T>>, Compare, Alloc)
-> flat_multimap<Key, T, Compare>;

template<class Key, class T, class Alloc>
flat_multimap(sorted_equivalent_t, initializer_list<pair<const Key, T>>, Alloc)
-> flat_multimap<Key, T>;
}

```

21.6.9.2 Constructors

[flatmultimap.cons]

```
flat_multimap(key_container_type key_cont, mapped_container_type mapped_cont);
```

1 *Effects:* Initializes `c.keys` with `std::move(key_cont)` and `c.values` with `std::move(mapped_cont)`; value-initializes `compare`; and sorts the range `[begin(), end())` with `compare`.

2 *Complexity:* Linear in N if the container arguments are already sorted as if with `compare` and otherwise $N \log N$, where N is `key_cont.size()`.

```
flat_multimap(sorted_equivalent_t, key_container_type key_cont,
              mapped_container_type mapped_cont);
```

3 *Effects:* Initializes `c.keys` with `std::move(key_cont)` and `c.values` with `std::move(mapped_cont)`; value-initializes `compare`.

4 *Complexity:* Constant.

```
template <class InputIterator>
flat_multimap(InputIterator first, InputIterator last, const key_compare& comp);
```

5 *Effects:* Initializes `compare` with `comp`, adds elements to `c` as if by:

```

    for (; first != last; ++first) {
        c.keys.insert(c.keys.end(), first->first);
        c.values.insert(c.values.end(), first->second);
    }

```

and finally sorts the range `[begin(),end())` with `compare`.

- 6 *Complexity:* Linear in N if the container arguments are already sorted as if with `compare` and otherwise $N \log N$, where N is `key_cont.size()`.

```

template <class InputIterator>
    flat_multimap(sorted_equivalent_t, InputIterator first, InputIterator last,
        const key_compare& comp = key_compare());

```

- 7 *Effects:* Initializes `compare` with `comp`, and adds elements to `c.keys` and `c.values` as if by:

```

    for (; first != last; ++first) {
        c.keys.insert(c.keys.end(), first->first);
        c.values.insert(c.values.end(), first->second);
    }

```

- 8 *Complexity:* Linear.

21.6.9.3 Constructors with allocators

[flatmultimap.cons.alloc]

```

template <class Alloc>
    flat_multimap(const key_compare& comp, const Alloc& a);

```

- 1 *Effects:* Initializes `compare` with `comp`, and performs uses-allocator construction (23.10.8.2) of both `c.keys` and `c.values` with `a`.

- 2 *Complexity:* Constant.

```

template <class InputIterator, class Alloc>
    flat_multimap(InputIterator first, InputIterator last,
        const key_compare& comp, const Alloc& a);

```

- 3 *Effects:* Initializes `compare` with `comp`, and performs uses-allocator construction (23.10.8.2) of both `c.keys` and `c.values` with `a`; adds elements to `c.keys` and `c.values` as if by:

```

    for (; first != last; ++first) {
        c.keys.insert(c.keys.end(), first->first);
        c.values.insert(c.values.end(), first->second);
    }

```

and finally sorts the range `[begin(),end())` with `compare`.

- 4 *Complexity:* Linear in N if the container arguments are already sorted as if with `compare` and otherwise $N \log N$, where N is `distance(first, last)`.

```

template <class InputIterator, class Alloc>
    flat_multimap(sorted_equivalent_t, InputIterator first, InputIterator last,
        const key_compare& comp, const Alloc& a);

```

- 5 *Effects:* Initializes `compare` with `comp`, and performs uses-allocator construction (23.10.8.2) of both `c.keys` and `c.values` with `a`; adds elements to `c.keys` and `c.values` as if by:

```

    for (; first != last; ++first) {
        c.keys.insert(c.keys.end(), first->first);
        c.values.insert(c.values.end(), first->second);
    }

```

- 6 *Complexity:* Linear.

21.6.9.4 Capacity

[flatmultimap.capacity]

```
size_type size() const noexcept;
```

1 *Effects:* Equivalent to: `return c.keys.size();`

```
size_type max_size() const noexcept;
```

2 *Effects:* Equivalent to: `return std::min<size_type>(c.keys.max_size(), c.values.max_size());`

21.6.9.5 Modifiers

[flatmultimap.modifiers]

```
flat_multimap& operator=(initializer_list<value_type> il);
```

1 *Effects:* Equivalent to:

```
clear();
insert(il);
```

```
template <class... Args> pair<iterator, bool> emplace(Args&&... args);
```

2 *Constraints:* `pair<key_type, mapped_type>(std::forward<Args>(args)...) is well-formed.`

3 *Effects:* First, constructs a `pair<key_type, value_type>` object `t` constructed with `std::forward<Args>(args)...`, then inserts `t` as if by:

```
auto key_it = upper_bound(c.keys.begin(), c.keys.end(), t.first);
auto value_it = c.values.begin() + distance(c.keys.begin(), key_it);
c.keys.emplace(key_it, std::move(t.first));
c.values.emplace(value_it, std::move(t.second));
```

4 *Returns:* An iterator that points to the inserted element.

```
template<class P> iterator insert(P&& x);
```

```
template<class P> iterator insert(const_iterator position, P&& x);
```

5 *Constraints:* `pair<key_type, mapped_type>(std::forward<P>(x)) is well-formed.`

6 *Effects:* The first form is equivalent to `return emplace(std::forward<P>(x)).` The second form is equivalent to `return emplace_hint(position, std::forward<P>(x)).`

```
template <class InputIterator>
```

```
void insert(sorted_equivalent_t, InputIterator first, InputIterator last);
```

7 *Expects:* The range `[first,last)` is sorted with respect to `compare`.

8 *Effects:* Equivalent to: `insert(first, last).`

9 *Complexity:* Linear.

```
void insert(sorted_equivalent_t, initializer_list<value_type> il);
```

10 *Effects:* Equivalent to `insert(sorted_equivalent_t, il.begin(), il.end()).`

```
void swap(flat_multimap& fm) noexcept;
```

11 *Constraints:* `is_nothrow_swappable_v<key_container_type> && is_nothrow_swappable_v<mapped_container_type> && is_nothrow_swappable_v<key_compare>` is true.

12 *Effects:* Equivalent to:

```
using std::swap;
swap(c.keys, fm.c.keys);
swap(c.values, fm.c.values);
swap(c.compare, fm.compare);
```

```
containers extract() &&;
```

Effects: Equivalent to:

```
containers temp;
temp.keys.swap(c.keys);
temp.values.swap(c.values);
return temp;
```

```
void replace(key_container_type&& key_cont, mapped_container_type&& mapped_cont);
```

13 *Expects:* `key_cont.size() == mapped_cont.size()` is true, and the elements of `key_cont` are sorted with respect to `compare`.

14 *Effects:* Equivalent to:

```
c.keys = std::move(key_cont);
c.values = std::move(mapped_cont);
```

21.6.9.6 Operators

[flatmultimap.ops]

```
template<class Key, class T, class Compare, class KeyContainer, class MappedContainer>
bool operator==(const flat_multimap<Key, T, Compare, KeyContainer, MappedContainer>& x,
                const flat_multimap<Key, T, Compare, KeyContainer, MappedContainer>& y);
```

1 *Effects:* Equivalent to: `return std::equal(x.begin(), x.end(), y.begin(), y.end());`

```
template<class Key, class T, class Compare, class KeyContainer, class MappedContainer>
bool operator!=(const flat_multimap<Key, T, Compare, KeyContainer, MappedContainer>& x,
                const flat_multimap<Key, T, Compare, KeyContainer, MappedContainer>& y);
```

2 *Returns:* `!(x == y)`.

```
template<class Key, class T, class Compare, class KeyContainer, class MappedContainer>
bool operator< (const flat_multimap<Key, T, Compare, KeyContainer, MappedContainer>& x,
                const flat_multimap<Key, T, Compare, KeyContainer, MappedContainer>& y);
```

3 *Effects:* Equivalent to: `return std::lexicographical_compare(x.begin(), x.end(), y.begin(), y.end());`

```
template<class Key, class T, class Compare, class KeyContainer, class MappedContainer>
bool operator> (const flat_multimap<Key, T, Compare, KeyContainer, MappedContainer>& x,
                const flat_multimap<Key, T, Compare, KeyContainer, MappedContainer>& y);
```

4 *Returns:* `y < x`.

```
template<class Key, class T, class Compare, class KeyContainer, class MappedContainer>
bool operator<=(const flat_multimap<Key, T, Compare, KeyContainer, MappedContainer>& x,
                const flat_multimap<Key, T, Compare, KeyContainer, MappedContainer>& y);
```

5 *Returns:* `!(y < x)`.

```
template<class Key, class T, class Compare, class KeyContainer, class MappedContainer>
bool operator>=(const flat_multimap<Key, T, Compare, KeyContainer, MappedContainer>& x,
                const flat_multimap<Key, T, Compare, KeyContainer, MappedContainer>& y);
```

6 *Returns:* `!(x < y)`.

21.6.9.7 Specialized algorithms**[flatmultimap.special]**

```
template<class Key, class T, class Compare, class KeyContainer, class MappedContainer>
    void swap(flat_multimap<Key, T, Compare, KeyContainer, MappedContainer>& x,
              flat_multimap<Key, T, Compare, KeyContainer, MappedContainer>& y) noexcept;
```

¹ *Constraints:* `is_nothrow_swappable_v<KeyContainer>` && `is_nothrow_swappable_v<MappedContainer>`
 && `is_nothrow_swappable_v<Compare>` is true.

² *Effects:* Equivalent to: `x.swap(y)`.

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